

Jonas Fischer

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Profile

AI Engineer and M.Sc. Informatics student specializing in Deep Learning, Computer Vision, and Robotics. Experience in designing, building, and deploying machine learning and GenAI systems in research and industry environments (TUM, BMW, Siemens, Fraunhofer). Strong focus on scalable ML architectures, RAG systems, and production-ready AI pipelines. Currently contributing to Large View Synthesis Models at the TUM Chair of Visual Computing and AI under Prof. Dr. Niessner.

Education

Technical University of Munich

Master of Science in Informatics; GPA: 1.3

Munich, Germany

Oct. 2024 – Present

- **Specialization:** Deep Learning, Computer Vision, Robotics
- **Relevant Coursework:** Introduction to Deep Learning, Advanced Deep Learning for Robotics, Advanced Deep Learning for Computer Vision

University of Freiburg

Bachelor of Science in Computer Science; GPA: 1.3

Freiburg, Germany

Oct. 2020 – Jun. 2024

- **Specialization:** Machine Learning and Robotics
- **Bachelor Thesis:** Parameterized and Dynamic Gaussian Mixture Models for Robot Manipulation (Grade 1.0)

Working Experience

BMW Group

Working Student – GenAI for Mainframe Migration

Munich, Germany

Jan. 2026 – Present

- **GenAI Engineering:** Built data processing and model-driven analysis pipelines to support the migration and transformation of legacy mainframe applications

Siemens Advanta

Working Student – Advanced Analytics and Artificial Intelligence

Munich, Germany

Oct. 2024 – Dec. 2025

- **Machine Learning:** Processed and clustered 70,000+ enterprise projects using OpenAI embeddings, UMAP, and K-means to uncover semantic structures and strategic thematic patterns
- **RAG Architecture:** Architected and deployed a scalable Retrieval-Augmented Generation (RAG) solution using Azure OpenAI, LangChain, and Microsoft Graph API for governance and compliance document retrieval
- **GenAI:** Designed and launched a GenAI-based contract analysis system for automated clause extraction and summarization, reducing manual contract review time by approximately 50%
- **Data Engineering:** Engineered SharePoint-to-Azure Search ETL pipelines with SQL-based monitoring to centralize and standardize document access

Freelancer

AI Engineer

Munich, Germany

Sept. 2025 – Present

- **GenAI Applications:** Architected and launched a customized RAG-based product recommendation system for Good Goods Germany GmbH, integrating LangGraph, Shopify API, and Python
- **Data Pipelines:** Systematized Shopify data synchronization workflows for reliable detection and propagation of product updates
- **Cloud Deployment:** Directed end-to-end Azure deployment including backend services, vector search infrastructure, and API integrations

Fraunhofer Institute for High-Speed Dynamics

Research Assistant

Freiburg, Germany

Jun. 2023 – Feb. 2024

- **Simulation Engineering:** Enhanced a Python-based satellite simulation framework by integrating orbital mechanics models and trajectory propagation algorithms
- **Scientific Analysis:** Applied numerical analysis with Pandas and NumPy to assess simulation accuracy and evaluate performance behavior

University of Freiburg

Student Assistant – Graph Theory

Freiburg, Germany

May 2024 – Sep. 2024

- **Teaching:** Directed weekly tutorials in Graph Theory, explaining formal proofs, algorithmic design, and complexity analysis to undergraduate students

University of Freiburg

Student Assistant – Logic for Computer Scientists

Freiburg, Germany

Oct. 2022 – Feb. 2023

- **Teaching:** Oversaw tutorials in propositional and predicate logic and assessed assignments and examinations

Selected Academic Projects

Self-Supervised Learning for Tactile Classification – *PyTorch, Google Cloud*

- Compared JEPA, MAE, and Autoencoder-based architectures for tactile perception tasks.
- Achieved a 3% increase in classification accuracy over a CNN baseline using predictive representation learning.
- Repository: github.com/JonasEtFi/SSL-for-Tactile-Classification

Reinforcement Learning for Smart Grids – *PyTorch, OpenAI Gymnasium, Weights & Biases*

- Executed and evaluated Pareto Conditioned Networks for multi-objective reinforcement learning.
- Modeled dynamic trade-offs between energy efficiency and operational safety within simulated smart grid environments.
- Repository: github.com/JonasEtFi/fork-commonpower-MORL

More projects: jonasetfi.github.io

Technical Skills

Programming: Python, SQL, C#

Machine Learning: PyTorch, Scikit-learn, XGBoost, Stable Baselines3

Computer Vision: Vision Transformers, Gaussian Splatting, Novel View Synthesis

GenAI / LLMs: LangChain, LangGraph, OpenAI API, RAG architectures, Embeddings

Cloud & MLOps: Azure, Google Cloud, Docker, Git, Weights & Biases

Data Engineering: NumPy, Pandas, Matplotlib, Seaborn, Flask, Microsoft Graph API

Additional Activities

TUMVision: Weekly discussion of state-of-the-art Computer Vision research papers

Sports: Field hockey (HC Wacker Munich), Hyrox

Hockey Coach: Player-coach of FT Freiburg men's hockey team (2023/2024 season)

Barista: Three years of professional hospitality experience

Munich, 22.3.2026